

Consider each matrix in Exercises 5 and 6 as the augmented matrix of a linear system. State in words the next two elementary row operations that should be performed in the process of solving the system.

5.
$$\left[\begin{array}{cccc|c} 1 & -4 & 5 & 0 & 7 \\ 0 & 1 & -3 & 0 & 6 \\ 0 & 0 & 1 & 0 & 2 \\ 0 & 0 & 0 & 1 & -5 \end{array} \right]$$
 Row 2 = Old Row 2 + 3 × Row 3

Row 1 = Old Row 1 - 5 × Row 3

$$\left[\begin{array}{cccc|c} 1 & -4 & 0 & 0 & -3 \\ 0 & 1 & 0 & 0 & 12 \\ 0 & 0 & 1 & 0 & 2 \\ 0 & 0 & 0 & 1 & -5 \end{array} \right]$$

In Exercises 7-10, the augmented matrix of a linear system has been reduced by row operations to the form shown. In each case, continue the appropriate row operations and describe the solution set of the original system.

7.
$$\left[\begin{array}{cccc|c} 1 & 7 & 3 & -4 & 3 \\ 0 & 1 & -1 & 3 & 1 \\ 0 & 0 & 0 & 1 & -2 \\ 0 & 0 & 1 & -2 & 1 \end{array} \right]$$
 Old row 2 = New row 2 + row 4

New row 1 = Old row 1 - 3 × row 4

$$\left[\begin{array}{cccc|c} 1 & 7 & 0 & 2 & 1 \\ 0 & 1 & 0 & 1 & 1 \\ 0 & 0 & 0 & 1 & -2 \\ 0 & 0 & 1 & -2 & 1 \end{array} \right]$$

New row 1 = Old row 1 - 7 × row 2

$$\left[\begin{array}{cccc|c} 1 & 0 & 0 & -5 & -5 \\ 0 & 1 & 0 & 1 & 1 \\ 0 & 0 & 0 & 1 & -2 \\ 0 & 0 & 1 & -2 & 1 \end{array} \right]$$

$$\begin{aligned} x_1 &= -5 \\ x_2 &= 1 \\ 0 &= 1 \\ x_3 &= -2 \end{aligned}$$

This is an inconsistent solution.